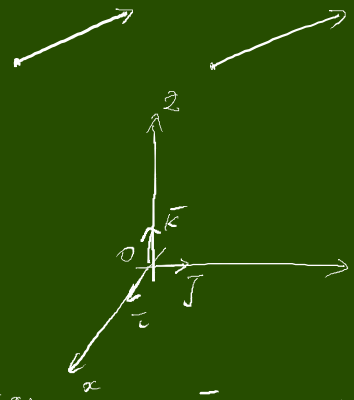


3D

$$\begin{cases} \bar{i} \times \bar{j} = \bar{k} = -\bar{j} \times \bar{i} \\ \bar{j} \times \bar{k} = \bar{i} = -\bar{k} \times \bar{j} \\ \bar{k} \times \bar{i} = \bar{j} = -\bar{i} \times \bar{k} \end{cases}$$

$$\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$$

$\bar{a} \times \bar{b}$



$$\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$$
$$\bar{b} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$$

$$\bar{a} \times \bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \dots$$

$(V, +, 0)$ grup abelian

$0_1, 0_2$

$$0_2 = 0_1 + 0_2 = 0_1$$

$$0_1 + 0_1 = 0_1$$
$$0_2 + 0_2 = 0_2$$

$0_k \cdot v = 0_V$?

$0_V \in V$

$0_k \in k$

$$0_k + 0_k = 0_k \quad / \cdot v$$

$$(0_k + 0_k) \cdot v = 0_k \cdot v$$

$$0_k \cdot v + 0_k \cdot v = 0_k \cdot v$$

$$\Rightarrow \quad 0_k \cdot v = 0_V$$

Exemple

1) $V = k^n$

$n = 1 \qquad v = (x) \qquad v = x$

$V = k^1 = k$

k este Sp. vect. peste el însuși

$V \quad k$

$k = \mathbb{R} \qquad \mathbb{R} = \mathbb{R}$ - Sp. vect

$k = \mathbb{C} \qquad \mathbb{C} = \mathbb{C}$ - Sp. vect

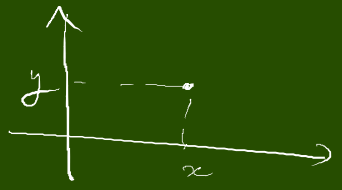
Obs. $\quad k \quad V$

$\alpha \in \mathbb{R} \quad z \in \mathbb{C} \quad z = x + iy \rightsquigarrow \begin{pmatrix} x \\ y \end{pmatrix}$

$\alpha(x + iy) = \alpha x + i(\alpha y)$

$V \quad k$

$\mathbb{C} = \mathbb{R}$ - Sp. vectorial



Q: $\mathbb{R} = \mathbb{C}$ - Sp. vect. ?

$V = (\mathbb{R}, +, \cdot)$

$\mathbb{C} \times \mathbb{R} \longrightarrow \mathbb{R}$?

$(z, \alpha) \longrightarrow z * \alpha \in \mathbb{R}$

$\in \mathbb{C} \quad \in \mathbb{R}$

$$\begin{cases} 1 * \alpha = \alpha \\ z_1 * (z_2 * \alpha) = (z_1 z_2) * \alpha \\ (z_1 + z_2) * \alpha = z_1 * \alpha + z_2 * \alpha \\ z * (\alpha + \beta) = z * \alpha + z * \beta \end{cases}$$

Exemple. 2) $V = M_{m,n}(k)$

k-sp. vect.

3) $V = k[X]$ $\quad n = \deg(P)$

$P(X) = a_0 + a_1 X + \dots + a_n X^n$ $\quad a_n \neq 0$

$a_0 \dots a_n \in k \quad X, X^2, \dots, X^n, \dots$

$Q(x) = b_0 + b_1 X + \dots + b_m X^m$

$P(x) + Q(x) = (a_0 + b_0) + (a_1 + b_1)X + \dots$

$P(x) = 2 + X$

$Q(x) = -3 + 2x - x^3$

$P(x) + Q(x) = -1 + 3x - x^3$

$\alpha \in k$, $P(x)$

$\alpha P(x) = (\alpha a_0) + (\alpha a_1) X + \dots + (\alpha a_n) X^n$

Thm. Bezout : $\alpha \in k \quad P \in K[x]$

$X - \alpha \mid P \iff P(\alpha) = 0$

$\Rightarrow$ $k[x]$ k-Sp.vect.

---

4) M $\quad \{f : M \xrightarrow[\text{fonction}]{} k\} = k^M$ not.

$k = \mathbb{R}$ $\quad M = \mathbb{R}$

$f, g : R \longrightarrow R$

$f + g : R \longrightarrow R$

$(f+g)(x) = f(x) + g(x)$

$\alpha f : R \longrightarrow R$

$(\alpha f)(x) = \alpha \cdot f(x)$

$\alpha \in R$

$\Rightarrow \quad R^R \quad R$ - sp. ve cf.

$U \subseteq V \qquad U \neq \emptyset \qquad v, w \in U$

$\lambda = \mu = 1_K$

$\lambda v + \mu w = v + w$

U subspaliu $\Rightarrow$ $v + w \in U$

$\mu = 0$

$\lambda v + \mu w = \lambda v \Rightarrow \lambda v \in U$

$0_V \in U$ da U è sottospazio

$\lambda v + \mu w = 0_V \in U$

$\lambda, \mu \in K$

$v, w \in U$

$\Rightarrow U \neq \emptyset$

$0_V \in U$ $\quad \{0_V\} \subseteq U \subseteq V$

Esempio $\quad V = \mathbb{R}^{\mathbb{R}}$

$U_1 = \{ f : \mathbb{R} \to \mathbb{R} \mid f \text{ cost} \}$

sottosp. vett.

$U_1 \supseteq U_2 = \{ f : \mathbb{R} \to \mathbb{R} \mid f \text{ polin.} \}$

$f(x) = a_0 + \dots + a_n x^n$

$$\underset{(m,n)(n,1)}{A \cdot x} = \underset{(m,1)}{O_m} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\{ x \mid Ax = O_m \} = \ker(A)$$

$$\forall \lambda, \mu \in k \quad \forall x, y \in \ker(A) \overset{?}{\Rightarrow}$$

$$\lambda x + \mu y \in \ker(A)$$

$$A(\lambda x + \mu y) = A(\lambda x) + A(\mu y)$$

$$= \lambda \underbrace{(Ax)}_{O_m} + \mu \underbrace{(Ay)}_{O_m}$$

$$= O_m$$

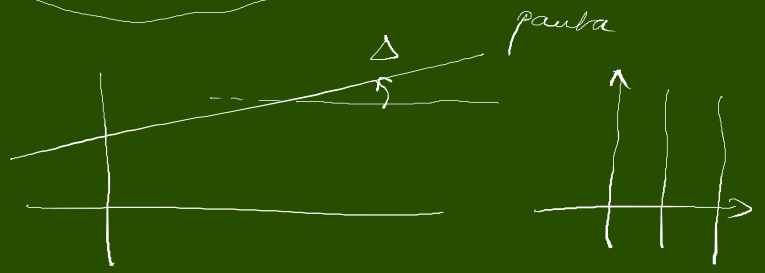
$A = O_{m,n}$ $\quad \ker(A) = k^n$

$m = n$ $\quad A = I_n$ $\quad \ker(A) = \{O_n\}$

2D : dreapta Δ $ax + by = c$ $b \neq 0$ $x = ct$

$c = 0$ $\Leftrightarrow$ $(0,0) \in \Delta$

$y = -\frac{a}{b}x + \frac{c}{b}$ $\quad y = mx + n$



$ax + by + c = 0$

$ax + by + cz + d = 0$ $\quad ax + by + c_0 = d$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

$$\begin{cases} x + 2y + 3z = 0 \\ 2x + 4y + 6z = 0 \quad | : 2 \end{cases}$$

$A \in M_{3,2}(\mathbb{R})$

$Ax = 0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$(3,2)(2,1)\ (3,1)$

$\ker(A) \subseteq \mathbb{R}^2$

$\operatorname{Im}(A) \subseteq \mathbb{R}^3$

Ax

$(3,2)(2,1)\ (3,1)$

?

?

$A \longrightarrow A^t \in M_{2,3}(\mathbb{R}$

$\ker(A^t) \subseteq \mathbb{R}^3$

$\operatorname{Im}(A^t) \subseteq \mathbb{R}^2$

$$A = \begin{pmatrix} 1 & -1 & 2 \\ -3 & 0 & 1 \end{pmatrix}$$

$\ker(A)$, ...

$\mathbb{R}^2 \longrightarrow$ grafic

$\mathbb{R}^3 \longrightarrow$ geogebra